

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industry based problem Solving	Assessment:	Assessment Tool 2 – Industry based problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze real-world business data using confidence intervals, hypothesis testing, test statistics, and p-value approaches to evaluate claims and support data-driven industry decisions.		
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Section / Batch:	AI-DS	Date:	18/08/26

Code of Conduct

I am B Tharun Kumar - 192424356 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *B Tharun Kumar*

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

		analyze the case deeply				
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Statistics Assessment — Confidence Intervals & Hypothesis Testing

Applied Industry Case Studies (Problems 6–10)

Problem 1 — Software Industry

A software company measures the response time of 100 API requests. The sample mean is 240 ms and standard deviation is 35 ms. Construct a 95% confidence interval for the average API response time.

Step-by-step Solution

1. Population and Sample:

Population = all API requests processed by the software system; Sample = 100 randomly selected API requests.

2. Parameter of Interest:

Population mean response time, μ (in milliseconds).

3. Point Estimate:

$$\bar{x} = 240 \text{ ms.}$$

4. Null Hypothesis (H_0):

Not applicable — no claimed benchmark value is given; the task only requires estimating μ via a confidence interval.

5. Alternative Hypothesis (H_1):

Not applicable (see above).

6. Significance Level (α):

$$\alpha = 0.05 \text{ (for 95\% confidence).}$$

7. Confidence Interval:

Computed below using the z-distribution since $n = 100 \geq 30$.

8. p-value:

Not applicable — no hypothesis test is being conducted.

9. Comparison with α : Not applicable.

Calculation

$$n = 100, \bar{x} = 240 \text{ ms}, s = 35 \text{ ms}, z(0.025) = 1.96$$

$$\text{Standard Error (SE)} = s / \sqrt{n} = 35 / \sqrt{100} = 3.5$$

$$\text{Margin of Error (ME)} = z \times \text{SE} = 1.96 \times 3.5 = 6.86$$

$$95\% \text{ CI} = \bar{x} \pm \text{ME} = 240 \pm 6.86 = (233.14 \text{ ms}, 246.86 \text{ ms})$$

Industry-Oriented Conclusion

We are 95% confident that the true average API response time for the entire system lies between 233.14 ms and 246.86 ms. This interval gives the engineering team a data-backed benchmark for monitoring performance, setting SLA thresholds, and detecting future degradation in service speed.

Problem 2 — E-commerce

An e-commerce platform finds that 420 out of 600 randomly selected visitors purchased a product. Construct a 95% confidence interval for the true conversion rate.

Step-by-step Solution

1. Population and Sample:

Population = all visitors to the e-commerce platform; Sample = 600 randomly selected visitors.

2. Parameter of Interest:

Population proportion, p (true conversion rate).

3. Point Estimate:

$$\hat{p} = 420 / 600 = 0.70 \text{ (70\%).}$$

4. Null Hypothesis (H_0):

Not applicable — no claimed conversion rate is given to test against; only estimation is required.

5. Alternative Hypothesis (H_1):

Not applicable (see above).

6. Significance Level (α):

$$\alpha = 0.05 \text{ (for 95\% confidence).}$$

7. Confidence Interval:

Computed below using the normal approximation for proportions.

8. p-value:

Not applicable — no hypothesis test is being conducted.

9. Comparison with α :

Not applicable.

Calculation

$$n = 600, x = 420, \hat{p} = 0.70, z(0.025) = 1.96$$

$$\text{Standard Error (SE)} = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.70 \times 0.30) / 600]} = \sqrt{0.00035} \approx 0.0187$$

$$\text{Margin of Error (ME)} = z \times \text{SE} = 1.96 \times 0.0187 \approx 0.0367$$

$$95\% \text{ CI} = \hat{p} \pm \text{ME} = 0.70 \pm 0.0367 = (0.6633, 0.7367), \text{ i.e. (66.33\%, 73.67\%)}$$

Industry-Oriented Conclusion

The platform can be 95% confident that the true conversion rate among all visitors lies between 66.33% and 73.67%. This range helps the marketing and product teams set realistic conversion targets, evaluate campaign effectiveness, and forecast revenue with statistical confidence.

Problem 3 — Telecom

A telecom company wants to estimate the average monthly data consumption of its users. A sample of 225 users has a mean consumption of 14.8 GB and standard deviation of 4.5 GB. Construct a 99% confidence interval for the population mean.

Step-by-step Solution

1. Population and Sample:

Population = all users of the telecom company; Sample = 225 randomly selected users.

2. Parameter of Interest:

Population mean monthly data consumption, μ (in GB).

3. Point Estimate:

$$\bar{x} = 14.8 \text{ GB.}$$

4. Null Hypothesis (H0):

Not applicable — no claimed consumption value is given; only estimation is required.

5. Alternative Hypothesis (H1): Not applicable (see above).

6. Significance Level (α):

$\alpha = 0.01$ (for 99% confidence).

7. Confidence Interval:

Computed below using the z-distribution since $n = 225 \geq 30$.

8. p-value:

Not applicable — no hypothesis test is being conducted.

9. Comparison with α :

Not applicable.

Calculation

$$n = 225, \bar{x} = 14.8 \text{ GB}, s = 4.5 \text{ GB}, z(0.005) = 2.576$$

$$\text{Standard Error (SE)} = s / \sqrt{n} = 4.5 / \sqrt{225} = 4.5 / 15 = 0.3$$

$$\text{Margin of Error (ME)} = z \times \text{SE} = 2.576 \times 0.3 = 0.7728$$

$$99\% \text{ CI} = \bar{x} \pm \text{ME} = 14.8 \pm 0.7728 = (14.0272 \text{ GB}, 15.5728 \text{ GB})$$

Industry-Oriented Conclusion

The telecom company can be 99% confident that the true average monthly data consumption across all users lies between 14.03 GB and 15.57 GB. This estimate supports network capacity planning, infrastructure investment decisions, and the design of appropriately sized data plans.

Problem 4 — Manufacturing

A machine historically produces components with a mean diameter of 50 mm. A sample of 36 components has a mean diameter of 50.8 mm and standard deviation of 2.4 mm. Using a 95% confidence interval, determine whether 50 mm is a plausible population mean.

Step-by-step Solution

1. Population and Sample:

Population = all components produced by the machine; Sample = 36 randomly selected components.

2. Parameter of Interest:

Population mean diameter, μ (in mm).

3. Point Estimate:

$$\bar{x} = 50.8 \text{ mm.}$$

4. Null Hypothesis (H0):

$\mu = 50 \text{ mm}$ (the machine's mean diameter has not changed from its historical value).

5. Alternative Hypothesis (H1):

$\mu \neq 50 \text{ mm}$ (the machine's mean diameter has shifted from the historical value) — two-tailed test.

6. Significance Level (α):

$$\alpha = 0.05 \text{ (for 95\% confidence).}$$

7. Confidence Interval & Test Statistic:

Computed below using the t-distribution with $df = 35$ (population SD unknown, sample size moderate).

8. p-value:

Two-tailed p-value corresponding to $t = 2.00$ with $df = 35 \approx 0.053$.

9. Comparison with α :

$$\text{p-value (0.053)} > \alpha (0.05) \rightarrow \text{fail to reject } H_0.$$

Calculation

$$n = 36, \bar{x} = 50.8 \text{ mm}, s = 2.4 \text{ mm}, df = 35, t(0.025, 35) \approx 2.030$$

$$\text{Standard Error (SE)} = s / \sqrt{n} = 2.4 / \sqrt{36} = 2.4 / 6 = 0.4$$

$$\text{Margin of Error (ME)} = t \times \text{SE} = 2.030 \times 0.4 = 0.812$$

$$95\% \text{ CI} = \bar{x} \pm \text{ME} = 50.8 \pm 0.812 = (49.988 \text{ mm}, 51.612 \text{ mm})$$

$$\text{Test statistic: } t = (\bar{x} - \mu_0) / \text{SE} = (50.8 - 50) / 0.4 = 2.00$$

Industry-Oriented Conclusion

Since the 95% confidence interval (49.988 mm, 51.612 mm) contains the historical value of 50 mm, and the p-value (≈ 0.053) is slightly greater than α (0.05), there is not enough statistical evidence to conclude that the machine's mean diameter has shifted. 50 mm remains a plausible population mean. However, since the result is very close to the threshold, the manufacturing team should continue monitoring the process for early signs of drift.

Problem 5— Banking

A bank claims that its average loan-processing time is 5 days. A sample of 49 applications has an average processing time of 5.6 days with standard deviation 1.4 days. Construct a 95% confidence interval and determine whether the company's claim is reasonable.

Step-by-step Solution

1. Population and Sample:

Population = all loan applications processed by the bank; Sample = 49 randomly selected applications.

2. Parameter of Interest:

Population mean loan-processing time, μ (in days).

3. Point Estimate:

$$\bar{x} = 5.6 \text{ days.}$$

4. Null Hypothesis (H0):

$\mu = 5$ days (the bank's claimed average processing time is correct).

5. Alternative Hypothesis (H1):

$\mu \neq 5$ days (the true average processing time differs from the claim) — two-tailed test.

6. Significance Level (α):

$$\alpha = 0.05 \text{ (for 95\% confidence).}$$

7. Confidence Interval & Test Statistic:

Computed below using the t-distribution with $df = 48$.

8. p-value:

Two-tailed p-value corresponding to $t = 3.00$ with $df = 48 \approx 0.0021$.

9. Comparison with α :

$$p\text{-value } (0.0021) < \alpha \text{ } (0.05) \rightarrow \text{reject } H_0.$$

Calculation

$$n = 49, \bar{x} = 5.6 \text{ days, } s = 1.4 \text{ days, } df = 48, t(0.025, 48) \approx 2.011$$

$$\text{Standard Error (SE)} = s / \sqrt{n} = 1.4 / \sqrt{49} = 1.4 / 7 = 0.2$$

$$\text{Margin of Error (ME)} = t \times SE = 2.011 \times 0.2 = 0.4022$$

$$95\% \text{ CI} = \bar{x} \pm ME = 5.6 \pm 0.4022 = (5.198 \text{ days, } 6.002 \text{ days})$$

$$\text{Test statistic: } t = (\bar{x} - \mu_0) / SE = (5.6 - 5) / 0.2 = 3.00$$

Industry-Oriented Conclusion

Since 5 days does not fall within the 95% confidence interval (5.198 days, 6.002 days), and the p-value (≈ 0.0021) is far smaller than α (0.05), we reject H_0 . There is strong statistical evidence that the bank's actual average loan-processing time is significantly longer than its claimed 5 days. The bank should review its internal process efficiency and revise its customer-facing turnaround-time claim accordingly.